

AI Plant Disease Detection System

Project Report

1. Introduction

Agriculture plays an important role in the global economy, and plant diseases are one of the major factors affecting crop production. Early identification of diseases helps farmers take preventive measures and reduce crop losses.

The **AI Plant Disease Detection System** is a deep learning-based application developed to automatically detect plant diseases from leaf images. The system uses **Transfer Learning with the ResNet50 architecture** to classify plant leaf images into different disease categories. The ResNet50 model was selected as the final deployment model due to its classification performance.

The application is developed using **TensorFlow, Keras, Streamlit, and Hugging Face Generative AI**. Users can upload a plant leaf image, and the system predicts the disease category along with AI-generated treatment recommendations.

2. Objectives

The main objectives of this project are:

- To develop an AI-based plant disease detection system using deep learning.
 - To classify plant leaf images into different disease categories.
 - To compare the performance of CNN and transfer learning models.
 - To improve classification performance using pre-trained deep learning architectures.
 - To provide disease treatment recommendations using Generative AI.
 - To develop an interactive user-friendly web application using Streamlit.
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3. Technologies Used

Programming Language

- Python

Deep Learning Framework

- TensorFlow
- Keras

Machine Learning Techniques

- Convolutional Neural Network (CNN)
- Transfer Learning

Deep Learning Models Used

- Custom CNN Model
- ResNet50
- EfficientNetB0
- MobileNetV2

Application Development

- Streamlit

Generative AI Integration

- Hugging Face Inference API

Data Processing Libraries

- NumPy
 - Pandas
 - PIL
 - Matplotlib
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4. Dataset Description

The project uses a plant leaf disease image dataset containing multiple classes of healthy and diseased plant leaves.

The dataset consists of:

- Different plant varieties
- Multiple disease categories
- Healthy leaf images
- Diseased leaf images

The images are preprocessed before training. Image resizing, normalization, and data augmentation techniques are applied to improve model performance.

5. Model Development

5.1 ResNet50 Model (Final Deployment Model)

The final plant disease detection system is based on the ResNet50 transfer learning architecture. ResNet50 is a deep convolutional neural network consisting of 50 layers and

uses residual connections to overcome the vanishing gradient problem. A pre-trained ResNet50 model trained on ImageNet was fine-tuned on the plant disease dataset to classify leaf images into multiple disease categories.

Then mention:

- Image Size: 224 × 224
- Preprocessing: ResNet50 preprocess_input()
- Output Layer: Softmax (38 classes)

5.2 Transfer Learning Models

Transfer learning models were implemented using pre-trained architectures.

ResNet50

ResNet50 is a deep neural network architecture that uses residual connections to solve the vanishing gradient problem. It enables efficient training of deeper networks.

EfficientNetB0

EfficientNetB0 uses optimized scaling methods to balance model accuracy and computational efficiency.

MobileNetV2

MobileNetV2 is a lightweight deep learning architecture designed for faster image classification with fewer computational resources.

6. Model Performance Evaluation

The performance of different models was evaluated based on accuracy.

Model	Accuracy
CNN	92.90%
MobileNetV2	92.43%
ResNet50	97.47%
EfficientNetB0	96.44%

Based on the evaluation results, the **ResNet50 transfer learning model achieved the highest accuracy and was selected as the final deployment model.**

7. Application Features

The developed Streamlit application provides the following features:

Plant Disease Prediction

- Upload a plant leaf image.
- Image preprocessing is performed automatically.
- The trained **ResNet50** deep learning model predicts the disease category.
- The detected disease name is displayed to the user.

AI-Based Treatment Recommendation

The system provides:

- Possible treatment methods.
- Disease prevention techniques.
- AI-generated agricultural recommendations.

User Interface

The application provides:

- Simple and interactive interface.
- Real-time prediction.
- Image display and prediction results.

Results:

Deploy

Navigation

Select Option

Home

Predict Disease

About

AI Plant Disease Detection System

Upload a plant leaf image to detect diseases using Deep Learning and receive AI-generated treatment recommendations.

ResNet50 Model Loaded Successfully

Total Disease Classes : 38

Welcome to AI Plant Disease Detection System

This application detects plant diseases using a ResNet50 Deep Learning model and provides AI-generated disease explanations using Hugging Face Large Language Models.

Features

Upload plant leaf image

ResNet50-based disease classification

Confidence score prediction

Hugging Face AI explanation

Treatment recommendations

Prevention tips

Deploy

Navigation

Select Option

Home

Predict Disease

About

AI Plant Disease Detection System

Upload a plant leaf image to detect diseases using Deep Learning and receive AI-generated treatment recommendations.

Upload Plant Leaf Image

Choose Leaf Image

Drag and drop file here

Limit 200MB per file • JPG, JPEG, PNG

Browse files

0b344c16-456c-4085-aa61-6a1c4cb93632__RS_HL 2538.JPG

16.8KB



Uploaded Leaf Image

Predict Disease

Prediction Completed

Predicted Disease

Blueberry__healthy

Confidence Score

Confidence: 99.86%

Navigation

Select Option

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- ☒ Predict Disease
- ☐ About

AI Disease Explanation

AI Recommendation Generated

Blueberry Leaf Spot Disease: Healthy Blueberries

1. Disease Overview

Blueberry leaf spot is a common disease that affects blueberry plants. It can cause significant damage to the leaves, which can affect the overall health and productivity of the plant.

2. Symptoms

- Brown Spots:** You'll see small, round, brown spots on the leaves. These spots can grow larger and merge together.
- Yellowing Leaves:** As the disease progresses, the leaves may turn yellow and fall off.
- Reduced Yield:** Infected plants may produce fewer berries or smaller berries.

3. Causes

- Fungal Spores:** The disease is caused by a fungus that thrives in humid conditions.
- Poor Air Circulation:** Plants that are crowded or have poor air circulation are more likely to get this disease.
- Overwatering:** Too much water can create a humid environment that the fungus loves.

4. Recommended Treatment

- Remove Infected Leaves:** Pick off and destroy any leaves that show signs of the disease.
- Prune Plants:** Thin out the branches to improve air circulation and sunlight penetration.
- Fungicides:** Use a fungicide specifically designed for blueberry leaf spot. Follow the instructions carefully.
- Water Management:** Water the plants at the base to keep the leaves dry. Avoid overhead watering.

5. Prevention Tips

- Choose Resistant Varieties:** Plant blueberry varieties that are resistant to leaf spot.
- Proper Spacing:** Space your plants far enough apart to allow good air circulation.
- Mulch:** Use a mulch around the base of the plants to help retain moisture and keep the soil cool.
- Regular Inspection:** Check your plants regularly for any signs of the disease and act quickly if you spot any.

By following these tips, you can help keep your blueberry plants healthy and productive!

Navigation

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- ☒ Predict Disease
- ☐ About



Model Information

Deployment Model: ResNet50

Models Compared:

- Custom CNN
- ResNet50
- EfficientNetB0
- MobileNetV2

Input Image Size: 224 × 224

Task: Multi-class Plant Disease Classification

Disease

Blueberry___healthy

Confidence

100.00%

AI Plant Disease Detection System | Deep Learning + Generative AI

8. Trained Model Files

Due to GitHub file size limitations, trained deep learning models are stored separately in Google Drive.

The model files include:

- CNN
- RestNet50
- EfficientNetB0

The trained model files can be downloaded from the following Google Drive link:

Download Models from Google Drive:

<https://drive.google.com/drive/folders/1zPc-vFmNNJCK-UeN3L2xAIO4ia-I5D6J?usp=sharing>

After downloading, the model files should be placed inside the project models folder.

Project structure:

Plant-Disease-Detection/

|

|— Screenshots/

|

| |— Home.png

|

| |— Upload Leaf.png

|

| |— Prediction Result.png

|

| |— AI Recommendation.png

|

| |— AI Recommendation 2.png

|

| |— Model Information.png

|

|— app.py

|— class_names.pkl

|— mobilenet_final.keras

|— HuggingFace.ipynb

|— Model Evaluation.ipynb

|— Plant.ipynb

|— Plant_Disease_Transfer_Learning_MobileNet_ResNet.ipynb

|— model_comparison.csv

```
|— requirements.txt
|— .gitignore
|— Report.docx
└— README.md
```

9. System Workflow

The working process of the system is:

1. User uploads a plant leaf image.
 2. The uploaded image is resized and preprocessed.
 3. The trained **ResNet50** model extracts deep image features and predicts the disease class.
 4. The model predicts the disease class.
 5. The predicted disease name is displayed.
 6. Generative AI provides treatment recommendations.
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10. Future Enhancements

The following improvements can be implemented in the future:

- Increase the dataset size for better accuracy.
 - Add more plant species and disease categories.
 - Deploy the system on cloud platforms.
 - Develop a mobile application for farmers.
 - Add multilingual support for better accessibility.
 - Integrate real-time field monitoring using IoT devices.
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11. Conclusion

The AI Plant Disease Detection System successfully demonstrates the application of deep learning and artificial intelligence in agriculture.

The system can automatically identify plant diseases from leaf images and provide useful recommendations. The comparison of CNN and transfer learning models demonstrated that

ResNet50 achieved the best performance for this dataset. The final application uses the ResNet50 model for disease prediction and integrates Hugging Face Generative AI to provide treatment recommendations, making it a practical solution for intelligent plant disease detection. This project helps reduce manual disease identification efforts and supports smart agriculture by providing an AI-powered solution for early disease detection.